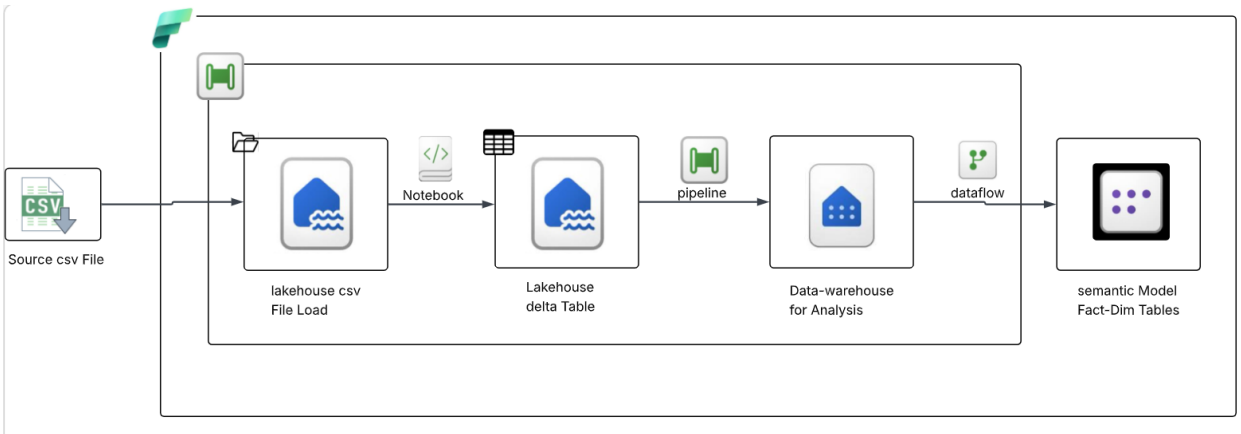
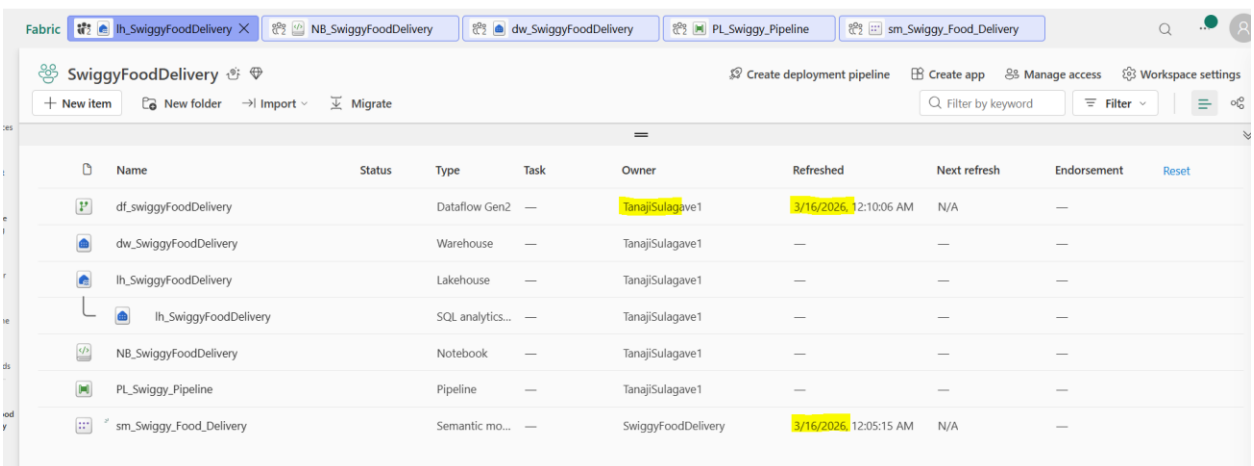
Microsoft Fabric Data Engineering Task

1) Architecture Diagram

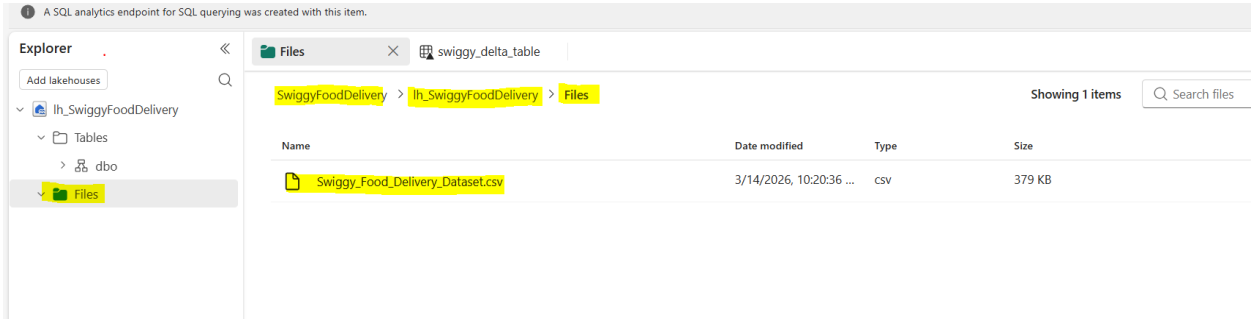


- All Fabric Services details:-

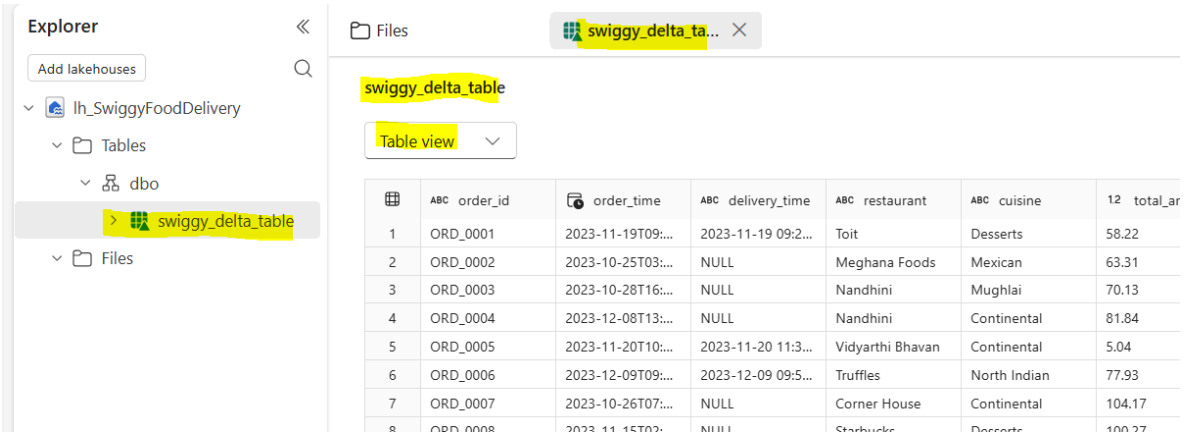


2) Lakehouse creation [Name:- **lh_SwiggyFoodDelivery**]

- CSV File upload:-



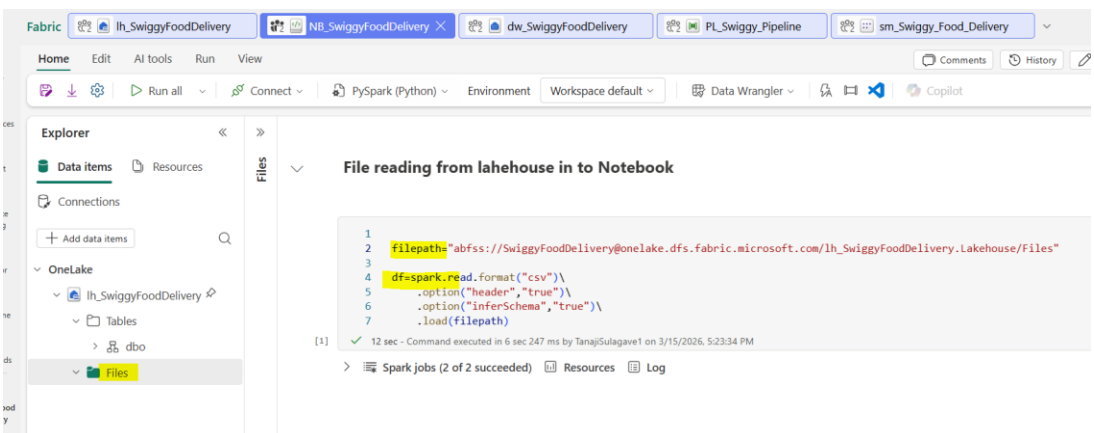
- Delta Table Creation:-



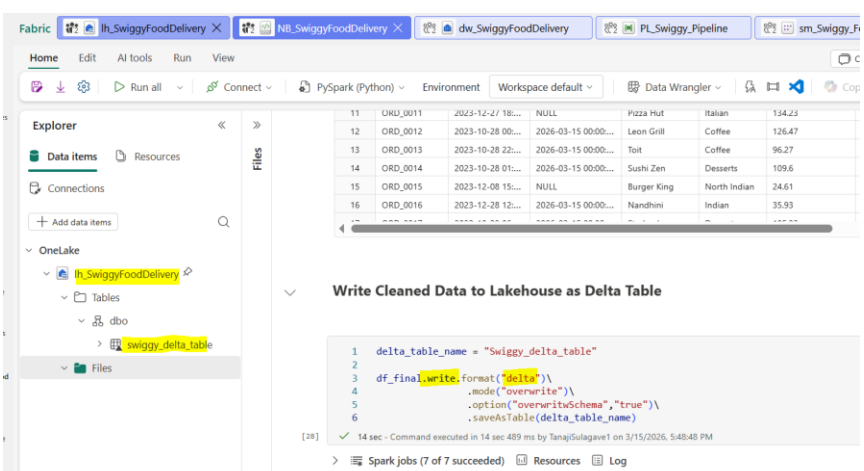
	order_id	order_time	delivery_time	restaurant	cuisine	total_ar
1	ORD_0001	2023-11-19T09:...	2023-11-19 09:2...	Toit	Desserts	58.22
2	ORD_0002	2023-10-25T03:...	NULL	Meghana Foods	Mexican	63.31
3	ORD_0003	2023-10-28T16:...	NULL	Nandhini	Mughlai	70.13
4	ORD_0004	2023-12-08T13:...	NULL	Nandhini	Continental	81.84
5	ORD_0005	2023-11-20T10:...	2023-11-20 11:3...	Vidyarthi Bhavan	Continental	5.04
6	ORD_0006	2023-12-09T09:...	2023-12-09 09:5...	Truffles	North Indian	77.93
7	ORD_0007	2023-10-26T07:...	NULL	Corner House	Continental	104.17

3) Notebook creation [Name:- **NB_SwiggyFoodDelivery**]

A. Source File connection:-



B. Load into Delta Table:-



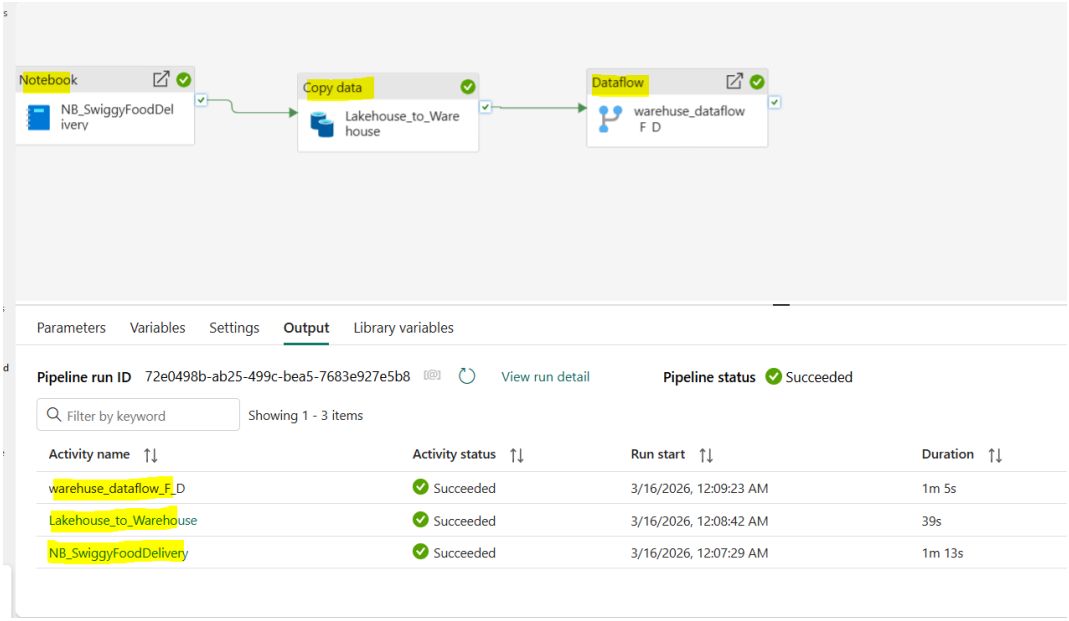
4. Pipeline creation [Name:- **PL_Swiggy_Pipeline**]

A. **Notebook Activity:-** In this activity, a notebook is executed to ingest CSV-formatted data. The notebook performs all the required transformations on the data and then loads the processed data into a **Delta table in the Lakehouse**.

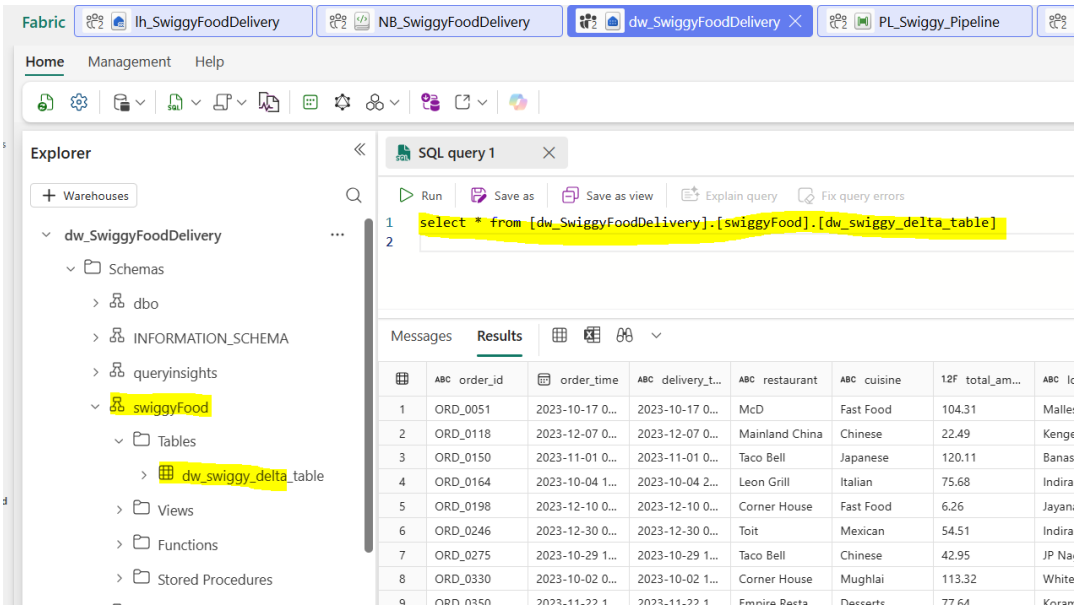
B. **Copy Activity:-** This activity copies data from the **Lakehouse Delta table** and loads it into the **Data Warehouse** under **swiggyFood** schema. The purpose of this step is to make the data available in the warehouse for further analysis and for writing SQL queries.

C. **Dataflow Activity:-** In this activity, **Dataflow Gen2** is executed. It retrieves data from the **Data Warehouse**, performs necessary transformations, and creates the required **Fact and Dimension tables**. The resulting data is then used to build and connect to the **Semantic Model**.

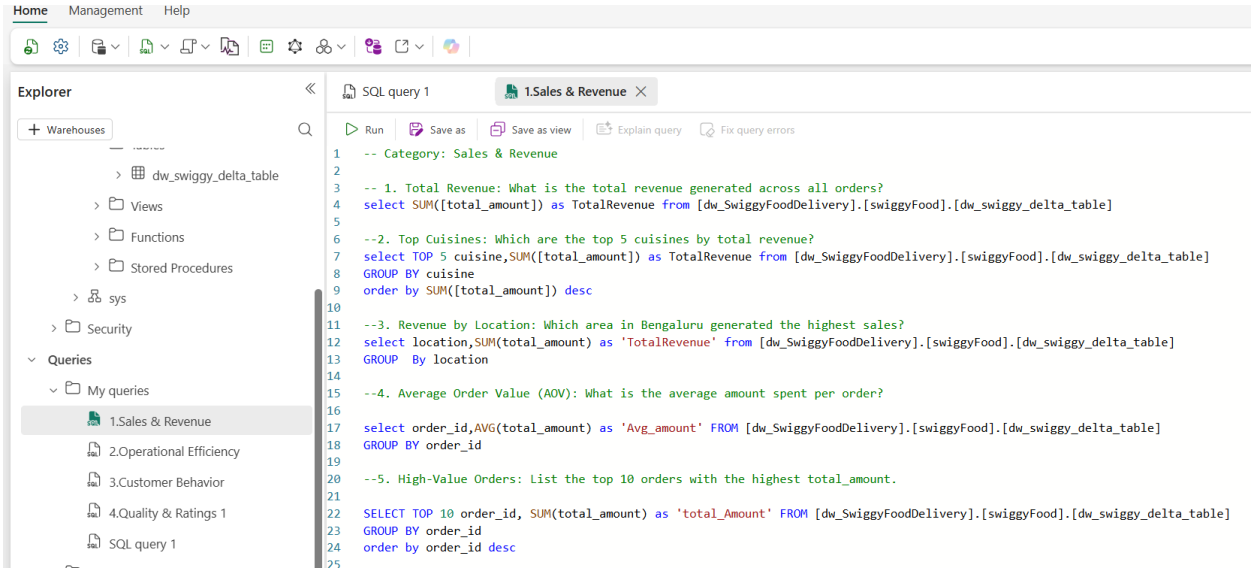
- Pipeline Execution Status:-



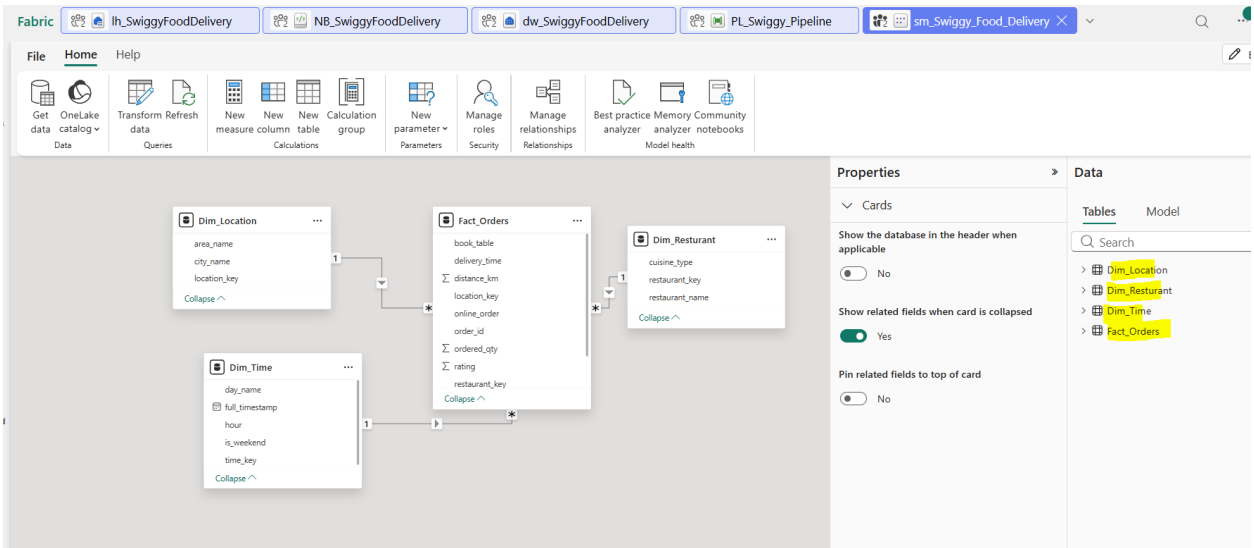
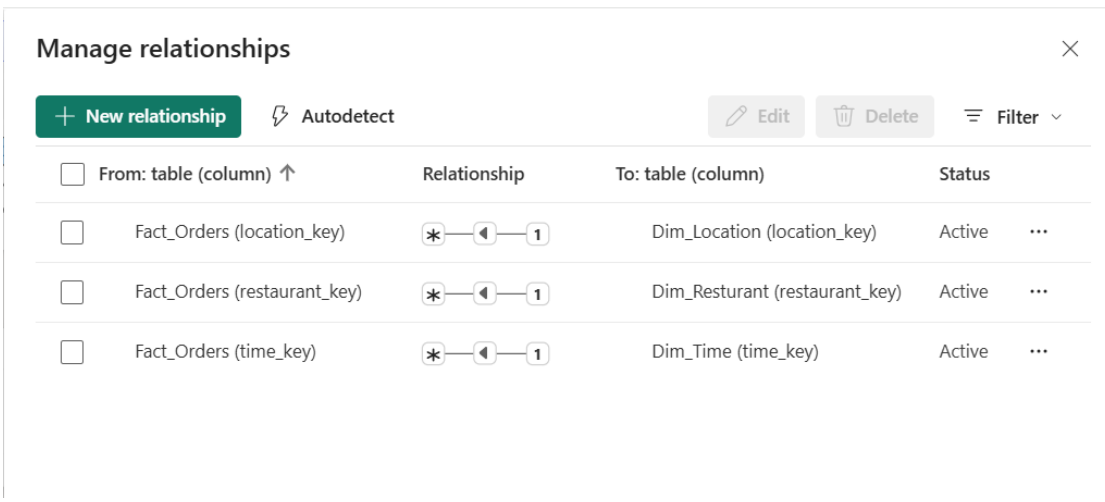
5. Warehouse creation [Name:- **dw_SwiggyFoodDelivery**]



- Data Analysis (Business requirements) :- Sample Snap



6. Create Sematic model [Name:- **sm_Swiggy_Food_Delivery**]



- <u>Relationships (Fact and Dim Table):-</u>